

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2024
Computer Networks (CS-317T)
Assignment # 1

Semester:V
Announced Date: 4-11-24
Total Marks: 3
Instructor Name: Razia Nisar Noorani

Batch: 2022F
Due Date: 8th Nov 24
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 1	Describe the key terminologies and technologies of computer networks.	PLO_1 (Academic Education)	C2 (Understanding)

Q1. Give an example of how layered architecture is implemented in a real-world networking scenario.

Imagine a Smart Traffic Management System that coordinates traffic lights, sensors, cameras, and connected vehicles to optimize urban traffic flow. Here’s how a layered architecture might be implemented:

- 1. Physical Layer:** This layer includes hardware like sensors, traffic lights, cameras, and other IoT devices embedded at intersections. It ensures connectivity between physical devices.
- 2. Data Link Layer:** The devices at each intersection form a local network, using a protocol like Ethernet or Wi-Fi to manage communication between local devices and aggregate data to a central system.
- 3. Network Layer:** Each intersection communicates with a central traffic control server using IP, which routes the data from different intersections across the citywide network, forming the backbone of the system.
- 4. Transport Layer:** The data from each intersection is sent as packets to ensure reliable, error-checked data transfer, using protocols like TCP for important data (traffic density information) or UDP for less critical updates (e.g., vehicle counts).
- 5. Application Layer:** The central server receives data, processes it, and applies machine learning algorithms to predict traffic flow. This layer also sends commands back to the intersections, adjusting signal timing or redirecting traffic based on real-time analysis.

This layered approach simplifies system management, isolating changes to specific layers—such as updating the machine learning algorithm (application layer) or upgrading intersection hardware (physical layer)—without affecting the whole system.

Q2: Describe the Internet Protocol Stack and its layers. How does it differ from the OSI model?

The Internet Protocol (IP) Stack, or TCP/IP model, has four layers that provide end-to-end data communication across the internet. Here’s a summary of each layer:

- 1. Application Layer:** This is where network applications operate (HTTP, FTP, DNS, etc.), allowing end-users to interact with the network and manage data transmission.
- 2. Transport Layer:** Responsible for data transport, this layer includes TCP (reliable, connection-oriented) and UDP (faster, connectionless), which handle data flow control and error-checking.
- 3. Internet Layer:** The Internet Layer, which uses IP, routes packets between devices across different networks, handling IP addressing and packet forwarding.
- 4. Link Layer:** The bottom layer, which includes physical network components and protocols, facilitates direct data transfer between connected devices, addressing physical addressing and MAC protocols.

Differences Between TCP/IP and OSI Models

The OSI Model has seven layers, while the TCP/IP Model has four. Here are key differences:

- **Layer Structure:** OSI's seven layers include Physical, Data Link, Network, Transport, Session, Presentation, and Application. TCP/IP combines the OSI's Physical and Data Link layers into the Link Layer and merges the Session, Presentation, and Application layers into one Application Layer.
- **Functionality:** The OSI model provides detailed modularity, while TCP/IP is more practical, focusing on the protocols commonly used in real-world networks.
- **Development History:** OSI was conceptualized as a theoretical framework, while TCP/IP emerged from practical network communication needs, driving its adoption on the Internet.

The TCP/IP model's simplicity and close alignment with real-world protocols make it more widely adopted in networking, despite the OSI model's detailed structure.

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